```
clear;
clc;
close all;

Fs = 10000;
fp = 1500;
fs = 2000;
N = 40;

% 1. Parks-McClellan (Equiripple) Design
f_bands = [0 fp fs Fs/2]/(Fs/2);
a_bands = [1 1 0 0];

b_pm = firpm(N,f_bands,a_bands);

% 2. Frequency-Sampling Design
b_fs = fir2(N,f_bands,a_bands);

% Display Numerical Outputs
fprintf('=== EXPERIMENT 5: PARKS-MCCLELLAN vs FREQUENCY SAMPLING ===\n\n');
```

```
=== EXPERIMENT 5: PARKS-MCCLELLAN vs FREQUENCY SAMPLING ===
```

```
fprintf('Filter Order (N): %d\n',N);
```

```
Filter Order (N): 40
```

```
fprintf('Passband Edge: %d Hz\n',fp);
```

```
Passband Edge: 1500 Hz
```

```
fprintf('Stopband Edge: %d Hz\n\n',fs);
```

```
Stopband Edge: 2000 Hz
```

```
fprintf('--- PARKS-MCCLELLAN COEFFICIENTS (First 10) ---\n');
```

```
--- PARKS-MCCLELLAN COEFFICIENTS (First 10) ---
```

```
disp(b_pm(1:10));
```

```
   -0.0015    0.0072    0.0053   -0.0010   -0.0081   -0.0072    0.0037    0.0142    0.0100   -0.0090
```

```
fprintf('--- FREQUENCY-SAMPLING COEFFICIENTS (First 10) ---\n');
```

```
--- FREQUENCY-SAMPLING COEFFICIENTS (First 10) ---
```

```
disp(b_fs(1:10));
```

```
    0.0000    0.0001    0.0002   -0.0001   -0.0008   -0.0009    0.0008    0.0035    0.0029   -0.0039
```

```
% Frequency Responses
[H_pm,f] = freqz(b_pm,1,1024,Fs);
[H_fs,~] = freqz(b_fs,1,1024,Fs);

% Magnitude Responses in dB
M_pm = 20*log10(max(abs(H_pm),1e-10));
M_fs = 20*log10(max(abs(H_fs),1e-10));

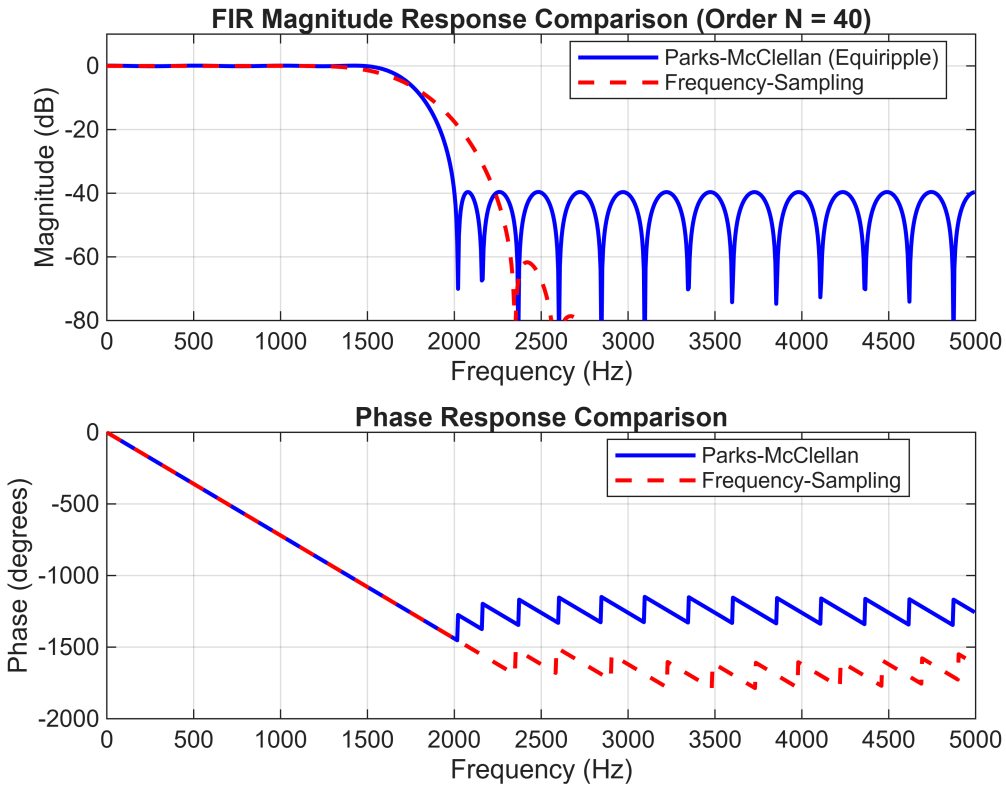
% Plotting
figure('Name','Parks-McClellan vs Frequency-Sampling');

subplot(2,1,1);
plot(f,M_pm,'b','LineWidth',1.5);
hold on;
plot(f,M_fs,'r--','LineWidth',1.5);
grid on;

legend('Parks-McClellan (Equiripple)', ...
    'Frequency-Sampling','Location','best');
title(['FIR Magnitude Response Comparison (Order N = ', ...
    num2str(N),')']);
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
ylim([-80 10]);

subplot(2,1,2);
plot(f,unwrap(angle(H_pm))*180/pi,'b','LineWidth',1.5);
hold on;
plot(f,unwrap(angle(H_fs))*180/pi,'r--','LineWidth',1.5);
grid on;

legend('Parks-McClellan','Frequency-Sampling', ...
    'Location','best');
title('Phase Response Comparison');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
```


FIR Magnitude Response Comparison (Order N = 40)
Parks-McClellan (Equiripple)
Frequency-Sampling
Magnitude (dB)
0
-20
-40
-60
-80
0
500
1000
1500
2000
2500
3000
3500
4000
4500
5000
Frequency (Hz)
Phase Response Comparison
Parks-McClellan
Frequency-Sampling
Phase (degrees)
0
-500
-1000
-1500
-2000
0
500
1000
1500
2000
2500
3000
3500
4000
4500
5000
Frequency (Hz)